

# VALIDATION REPORT

## OPERATIONAL QUALIFICATION

### SHELF LIFE & DEGRADATION ANALYSIS MODULE

**JR Validated Environment — Shelf Life & Degradation Analysis Module v2.0**

| Field            | Value                                                                                   |
|------------------|-----------------------------------------------------------------------------------------|
| Document Number  | JR-VR-SHELF-001                                                                         |
| Title            | Validation Report — Operational Qualification, Shelf Life & Degradation Analysis Module |
| Validation Plan  | JR-VP-SHELF-001 v2.0                                                                    |
| System           | JR Validated Environment — Shelf Life Module                                            |
| Module Version   | 2.0                                                                                     |
| Document Version | 3.0                                                                                     |
| Execution Date   | 2026-10-01                                                                              |
| OQ Result        | PASS — 77 / 77 tests passed, 0 failures, 0 deviations                                   |
| Author           | Joep Rous                                                                               |
| Reviewer         | [Name]                                                                                  |
| Approver         | [Name]                                                                                  |

## Version History

| Version | Date       | Author    | Description                                                                                                                                                                       |
|---------|------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | 2026-04-19 | Joep Rous | Initial OQ execution report. All 57 test cases passed. No deviations.                                                                                                             |
| 2.0     | 2026-05-18 | Joep Rous | Updated for Shelf Life module v2.0. Added --report DV report TCs for all 5 scripts (Q10-011..013, ARR-011..013, LIN-017..019, POOL-011..013, EXT-012..014). 72 / 72 tests passed. |
| 3.0     | 2026-10-01 | Joep Rous | jrscripsts v5.0.0, code review 2026-10: re-execution of the full suite against the revised validation plan. 77 / 77 tests passed.                                                 |

## Approval Signatures

| Role          | Name      | Signature | Date       |
|---------------|-----------|-----------|------------|
| Author        | Joep Rous |           | 2026-10-01 |
| Reviewer      | [Name]    |           |            |
| Approver (QA) | [Name]    |           |            |

## 1. Purpose

This document reports the execution and outcome of the Operational Qualification (OQ) test suite for the JR Validated Environment Shelf Life & Degradation Analysis Module, version 2.0. It provides objective evidence that all Shelf Life scripts perform as specified in OQ Validation Plan JR-VP-SHELF-001 v2.0 under the defined test conditions.

This report records the test environment, execution results for all 77 test cases, and the formal OQ conclusion. Results are derived directly from the evidence file generated by admin\_shelf\_life\_oq.

## 2. Scope

### 2.1 In Scope

- All 5 R scripts in repos/shelf\_life/R/ (jrc\_shelf\_life\_q10, jrc\_shelf\_life\_arrhenius, jrc\_shelf\_life\_linear, jrc\_shelf\_life\_poolability, jrc\_shelf\_life\_extrapolate)
- Correct computation of AF and real-time equivalent via Q10 (ASTM F1980) and Arrhenius (ISO 11607 / ICH Q1E) methods; numerical correctness verified by independent Python arithmetic
- Correct linear degradation modelling: Brown-Forsythe test, OLS regression, shelf life estimation as CI of predicted mean crossing the specification limit (ICH Q1E)
- Correct --transform log behaviour: log-scale OLS fit with CI back-transformed via exp()
- Correct ICH Q1E batch poolability: two-step ANCOVA at  $\alpha=0.25$ ; FULL/PARTIAL/DO NOT POOL decision
- Correct extrapolation with transform-aware CI back-transformation; ICH Q1E 50% warning and 100% hard stop
- Correct rejection of invalid inputs; bypass-protection verification (RENV\_PATHS\_ROOT check)
- PNG and model CSV output to ~/Downloads/

### 2.2 Out of Scope

- Infrastructure scripts in bin/ and admin/ (covered by JR-IQ-001)
- Performance qualification (PQ) with real product stability data
- Validation of R interpreter and third-party packages (ggplot2, nlme)
- Statistical validation of the underlying methods (Arrhenius kinetics, ICH Q1E poolability rules, OLS inference) — see JR-VP-SHELF-001 Section 3 for references

## 3. Reference Documents

| Document ID          | Title                                                          | Location                                               |
|----------------------|----------------------------------------------------------------|--------------------------------------------------------|
| JR-VP-SHELF-001 v2.0 | Validation Plan — OQ, Shelf Life & Degradation Analysis Module | repos/shelf_life/docs/shelf_life_validation_plan.pdf   |
| JR-VP-001 v1.0       | Validation Plan — Installation Qualification                   | docs/                                                  |
| JR-IQ-001            | IQ Execution Evidence                                          | ~/jrscript/<PROJECT_ID>/validation/IQ_validation_*.txt |
| OQ Evidence          | Shelf Life OQ Execution Evidence (pytest output + header)      | shelf_life_oq_execution_20261001T171453.txt            |

## 4. Test Environment

| Component             | Details                                      |
|-----------------------|----------------------------------------------|
| Execution date / time | 2026-10-01T17:14:53                          |
| Host                  | Joeps-MacBook-Air.local                      |
| Operating System      | 26.6.2                                       |
| R Version             | 4.6.0                                        |
| Python Version        | 3.11.9                                       |
| pytest Version        | pytest 8.3.4                                 |
| Test runner           | repos/shelf_life/admin_shelf_life_oq         |
| Test file             | repos/shelf_life/oq/test_shelf_life_suite.py |
| Test data             | repos/shelf_life/oq/data/ (13 CSV fixtures)  |
| Execution command     | bash repos/shelf_life/admin_shelf_life_oq    |
| OQ evidence file      | shelf_life_oq_execution_20261001T171453.txt  |

## 5. Test Results

All 77 test cases were executed on 2026-10-01. Results are derived from the OQ evidence file and reproduced below. The evidence file contains the complete pytest output including individual test IDs, pass/fail status, and timing.

| Script                     | Tests Executed | Result     |
|----------------------------|----------------|------------|
| jrc_shelf_life_q10         | 13             | 13/13 PASS |
| jrc_shelf_life_arrhenius   | 13             | 13/13 PASS |
| jrc_shelf_life_linear      | 24             | 24/24 PASS |
| jrc_shelf_life_poolability | 13             | 13/13 PASS |
| jrc_shelf_life_extrapolate | 14             | 14/14 PASS |
| TOTAL                      | 77             | 77/77 PASS |

### 5.1 jrc\_shelf\_life\_q10 — 13/13 PASS

| TC ID            | Description                                                          | Expected                                            | Result |
|------------------|----------------------------------------------------------------------|-----------------------------------------------------|--------|
| TC-SHELF-Q10-001 | Valid inputs                                                         | exit 0; key sections in output                      | PASS   |
| TC-SHELF-Q10-002 | Known AF = 8.0 ( $\pm 0.001$ ), real_time = 208.0 ( $\pm 0.01$ )     | Independent Python arithmetic reference             | PASS   |
| TC-SHELF-Q10-003 | Sensitivity table rows Q10 -0.5, Q10, Q10+0.5                        | exit 0; '1.5', 'stated value', '2.5' in output      | PASS   |
| TC-SHELF-Q10-004 | Q10=1.2 out-of-range warning                                         | exit 0; range warning in output                     | PASS   |
| TC-SHELF-Q10-005 | accel_temp $\leq$ real_temp rejected                                 | exit non-zero                                       | PASS   |
| TC-SHELF-Q10-006 | No arguments                                                         | exit non-zero or usage                              | PASS   |
| TC-SHELF-Q10-007 | Non-numeric q10                                                      | exit non-zero                                       | PASS   |
| TC-SHELF-Q10-008 | Negative accel_time                                                  | exit non-zero                                       | PASS   |
| TC-SHELF-Q10-009 | --help flag                                                          | exit 0; 'Usage' in output                           | PASS   |
| TC-SHELF-Q10-010 | Bypass protection                                                    | exit non-zero; RENV_PATHS_ROOT in output            | PASS   |
| TC-SHELF-Q10-011 | --report $\rightarrow$ *_q10_dv_report.docx or .html in ~/Downloads/ | exit 0; .docx or .html file found with recent mtime | PASS   |
| TC-SHELF-Q10-012 | --report JSON sidecar *_q10_dv_report_data.json (or .docx accepted)  | exit 0; .docx or JSON found with recent mtime       | PASS   |

|                  |                                                           |                                                                        |      |
|------------------|-----------------------------------------------------------|------------------------------------------------------------------------|------|
| TC-SHELF-Q10-013 | --report JSON: report_type=="dv",<br>verdict_pass is True | exit 0; if JSON present:<br>report_type=="dv", verdict_pass is<br>True | PASS |
|------------------|-----------------------------------------------------------|------------------------------------------------------------------------|------|

## 5.2 jrc\_shelf\_life\_arrhenius — 13/13 PASS

| TC ID            | Description                                                                      | Expected                                                               | Result |
|------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------|--------|
| TC-SHELF-ARR-001 | Valid inputs                                                                     | exit 0; key sections in output                                         | PASS   |
| TC-SHELF-ARR-002 | Known AF = 13.7826 ( $\pm 0.001$ ),<br>real_time = 358.35 ( $\pm 0.05$ )         | Independent math.exp() reference                                       | PASS   |
| TC-SHELF-ARR-003 | Kelvin inputs give same AF as<br>Celsius                                         | AF matches $\pm 0.01$                                                  | PASS   |
| TC-SHELF-ARR-004 | Sensitivity table Ea-2, Ea, Ea+2                                                 | exit 0; all three values in output                                     | PASS   |
| TC-SHELF-ARR-005 | Ea=30 out-of-range warning                                                       | exit 0; range warning in output                                        | PASS   |
| TC-SHELF-ARR-006 | accel_temp $\leq$ real_temp rejected                                             | exit non-zero                                                          | PASS   |
| TC-SHELF-ARR-007 | No arguments                                                                     | exit non-zero or usage                                                 | PASS   |
| TC-SHELF-ARR-008 | Invalid --unit F                                                                 | exit non-zero                                                          | PASS   |
| TC-SHELF-ARR-009 | --help flag                                                                      | exit 0; 'Usage' in output                                              | PASS   |
| TC-SHELF-ARR-010 | Bypass protection                                                                | exit non-zero; RENV_PATHS_ROOT in<br>output                            | PASS   |
| TC-SHELF-ARR-011 | --report $\rightarrow$<br>*_arrhenius_dv_report.docx or<br>.html in ~/Downloads/ | exit 0; .docx or .html file found with<br>recent mtime                 | PASS   |
| TC-SHELF-ARR-012 | --report JSON sidecar<br>*_arrhenius_dv_report_data.json<br>(or .docx accepted)  | exit 0; .docx or JSON found with<br>recent mtime                       | PASS   |
| TC-SHELF-ARR-013 | --report JSON: report_type=="dv",<br>verdict_pass is True                        | exit 0; if JSON present:<br>report_type=="dv", verdict_pass is<br>True | PASS   |

## 5.3 jrc\_shelf\_life\_linear — 24/24 PASS

| TC ID            | Description                                                    | Expected                                        | Result |
|------------------|----------------------------------------------------------------|-------------------------------------------------|--------|
| TC-SHELF-LIN-001 | Valid 15-row dataset                                           | exit 0; key sections in output                  | PASS   |
| TC-SHELF-LIN-002 | Known shelf life = 24.871 $\pm 0.05$ (one-<br>sided 95% bound) | Independent Python OLS + bisection<br>reference | PASS   |
| TC-SHELF-LIN-003 | Homogeneous variance — BF pass                                 | exit 0; 'homogeneous' in output                 | PASS   |
| TC-SHELF-LIN-004 | Heterogeneous variance — warning,<br>exit 0                    | warning in output                               | PASS   |
| TC-SHELF-LIN-005 | PNG saved to ~/Downloads/                                      | *_jrc_shelf_life_linear.png found               | PASS   |
| TC-SHELF-LIN-006 | Model CSV saved to ~/Downloads/                                | *_jrc_shelf_life_linear_model.csv<br>found      | PASS   |
| TC-SHELF-LIN-007 | --direction high                                               | exit 0; 'upper' or 'high' in output             | PASS   |
| TC-SHELF-LIN-008 | Values below spec at t=0<br>(direction=low)                    | exit non-zero                                   | PASS   |
| TC-SHELF-LIN-009 | No arguments                                                   | exit non-zero or usage                          | PASS   |
| TC-SHELF-LIN-010 | Missing 'value' column                                         | exit non-zero; 'value' in error                 | PASS   |
| TC-SHELF-LIN-011 | < 3 distinct time points                                       | exit non-zero                                   | PASS   |
| TC-SHELF-LIN-    | Bypass protection                                              | exit non-zero; RENV_PATHS_ROOT in               | PASS   |

|                  |                                                                                        |                                                                     |      |
|------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------|------|
| 012              |                                                                                        | output                                                              |      |
| TC-SHELF-LIN-013 | --transform log — happy path                                                           | exit 0; 'log' in output                                             | PASS |
| TC-SHELF-LIN-014 | --transform log — known shelf life ( $\pm 0.05$ )                                      | Independent Python log-OLS + bisection reference                    | PASS |
| TC-SHELF-LIN-015 | --transform log with value $\leq 0$                                                    | exit non-zero; positive/ $\leq 0$ in error                          | PASS |
| TC-SHELF-LIN-016 | --transform log — model CSV transform field                                            | model CSV contains transform = log                                  | PASS |
| TC-SHELF-LIN-017 | --report →<br>*_shelf_life_linear_dv_report.docx or<br>.html in ~/Downloads/           | exit 0; .docx or .html file found with recent mtime                 | PASS |
| TC-SHELF-LIN-018 | --report JSON sidcar<br>*_shelf_life_linear_dv_report_data.json<br>(or .docx accepted) | exit 0; .docx or JSON found with recent mtime                       | PASS |
| TC-SHELF-LIN-019 | --report JSON: report_type=="dv",<br>verdict_pass is True (homogeneous dataset)        | exit 0; if JSON present:<br>report_type=="dv", verdict_pass is True | PASS |
| TC-SHELF-LIN-020 | Verdict PASS: shelf life 24.87 $\geq$ t_max 24                                         | exit 0; 'Verdict: PASS'                                             | PASS |
| TC-SHELF-LIN-021 | Batch column, auto → FULL POOL                                                         | exit 0; 'FULL POOL'                                                 | PASS |
| TC-SHELF-LIN-022 | Batch column, auto → PARTIAL POOL,<br>limiting batch C 1.1610                          | exit 0; 1.1610                                                      | PASS |
| TC-SHELF-LIN-023 | Batch column, auto → DO NOT POOL,<br>limiting batch C 3.9513                           | exit 0; 3.9513                                                      | PASS |
| TC-SHELF-LIN-024 | --pool without batch column → error                                                    | exit non-zero                                                       | PASS |

## 5.4 jrc\_shelf\_life\_poolability — 13/13 PASS

| TC ID             | Description                                                            | Expected                                            | Result |
|-------------------|------------------------------------------------------------------------|-----------------------------------------------------|--------|
| TC-SHELF-POOL-001 | Valid 3-batch dataset                                                  | exit 0; 'Decision', 'POOL', Step 1/2 in output      | PASS   |
| TC-SHELF-POOL-002 | FULL POOL decision; both ANCOVA steps p > 0.25                         | exit 0; 'FULL POOL'; p1 > 0.25; p2 > 0.25           | PASS   |
| TC-SHELF-POOL-003 | DO NOT POOL; Step 1 p < 0.05                                           | exit 0; 'DO NOT POOL'; Step 1 p < 0.05              | PASS   |
| TC-SHELF-POOL-004 | PARTIAL POOL; Step 1 p > 0.25, Step 2 p < 0.01                         | exit 0; 'PARTIAL POOL'; p1 > 0.25; p2 < 0.01        | PASS   |
| TC-SHELF-POOL-005 | PNG saved to ~/Downloads/                                              | *_jrc_shelf_life_poolability.png found              | PASS   |
| TC-SHELF-POOL-006 | No arguments                                                           | exit non-zero or usage                              | PASS   |
| TC-SHELF-POOL-007 | Missing 'batch' column                                                 | exit non-zero; 'batch' in error                     | PASS   |
| TC-SHELF-POOL-008 | Only 1 batch                                                           | exit non-zero                                       | PASS   |
| TC-SHELF-POOL-009 | < 3 time points per batch                                              | exit non-zero                                       | PASS   |
| TC-SHELF-POOL-010 | Bypass protection                                                      | exit non-zero; RENV_PATHS_ROOT in output            | PASS   |
| TC-SHELF-POOL-011 | --report →<br>*_poolability_dv_report.docx or<br>.html in ~/Downloads/ | exit 0; .docx or .html file found with recent mtime | PASS   |
| TC-SHELF-POOL-012 | --report JSON sidcar<br>*_poolability_dv_report_data.json              | exit 0; .docx or JSON found with recent mtime       | PASS   |

|                   |                                                                           |                                                                  |      |
|-------------------|---------------------------------------------------------------------------|------------------------------------------------------------------|------|
|                   | (or .docx accepted)                                                       |                                                                  |      |
| TC-SHELF-POOL-013 | --report JSON: report_type=="dv", verdict_pass is True (poolable dataset) | exit 0; if JSON present: report_type=="dv", verdict_pass is True | PASS |

## 5.5 jrc\_shelf\_life\_extrapolate — 14/14 PASS

| TC ID            | Description                                                                      | Expected                                                         | Result |
|------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------|--------|
| TC-SHELF-EXT-001 | Valid model CSV, target within range                                             | exit 0; 'Projection', 'Fitted value'                             | PASS   |
| TC-SHELF-EXT-002 | Known fit = 84.502 ( $\pm 0.01$ ), one-sided lower bound = 84.071 ( $\pm 0.05$ ) | Closed-form from known coefficients + NIST t_crit                | PASS   |
| TC-SHELF-EXT-003 | Target within range, above spec — PASS                                           | exit 0; 'supported' in output                                    | PASS   |
| TC-SHELF-EXT-004 | Target beyond shelf life — FAIL message                                          | exit 0; 'NOT supported' or 'crossed'                             | PASS   |
| TC-SHELF-EXT-005 | >50% extrapolation warning                                                       | exit 0; warning in output                                        | PASS   |
| TC-SHELF-EXT-006 | >100% extrapolation hard stop                                                    | exit non-zero; 'rejected' or '100' in output                     | PASS   |
| TC-SHELF-EXT-007 | No arguments                                                                     | exit non-zero or usage                                           | PASS   |
| TC-SHELF-EXT-008 | Missing model file                                                               | exit non-zero; 'not found' in output                             | PASS   |
| TC-SHELF-EXT-009 | Wrong-script model file                                                          | exit non-zero; 'script' in error                                 | PASS   |
| TC-SHELF-EXT-010 | Negative target_time                                                             | exit non-zero                                                    | PASS   |
| TC-SHELF-EXT-011 | Bypass protection                                                                | exit non-zero; RENV_PATHS_ROOT in output                         | PASS   |
| TC-SHELF-EXT-012 | --report → *_extrapolate_dv_report.docx or .html in ~/Downloads/                 | exit 0; .docx or .html file found with recent mtime              | PASS   |
| TC-SHELF-EXT-013 | --report JSON sidecar *_extrapolate_dv_report_data.json (or .docx accepted)      | exit 0; .docx or JSON found with recent mtime                    | PASS   |
| TC-SHELF-EXT-014 | --report JSON: report_type=="dv", verdict_pass is True (CI > spec at target=20)  | exit 0; if JSON present: report_type=="dv", verdict_pass is True | PASS   |

## 6. Deviations

No deviations were identified during OQ execution. All 77 test cases passed.

## 7. OQ Conclusion

The Operational Qualification of the JR Validated Environment Shelf Life & Degradation Analysis Module v2.0 is complete.

**Result: PASS**

All 77 automated test cases specified in JR-VP-SHELF-001 v2.0 were executed on 2026-10-01 and 77 passed. Results are derived from the OQ evidence file listed below.

- jrc\_shelf\_life\_q10 — Accelerated Ageing via Q10 Method (ASTM F1980)
- jrc\_shelf\_life\_arrhenius — Accelerated Ageing via Arrhenius Kinetics (ISO 11607 / ICH Q1E)
- jrc\_shelf\_life\_linear — Linear Degradation Model with Shelf Life Estimation (ICH Q1E)

- jrc\_shelf\_life\_poolability — ICH Q1E Batch Poolability Analysis (Two-Step ANCOVA)
- jrc\_shelf\_life\_extrapolate — CI Projection to Target Time with Transform Support

*Evidence of test execution is archived at: shelf\_life\_oq\_execution\_20261001T171453.txt*

The Shelf Life & Degradation Analysis module is released as version 2.0 of repos/shelf\_life/. Any subsequent change to the scripts, wrappers, or help files within repos/shelf\_life/ will require a change assessment and, if the change affects functionality covered by this OQ, re-execution of the affected test cases.